



Unit 3 · Outcome 1 · Maintaining Biodiversity

Why genetic diversity matters when the environment changes

KK02

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 02 Why genetic diversity matters when the environment changes A population can only survive a new threat using alleles it already carries

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



Why genetic diversity matters when the environment changes

A population can only survive a new threat using alleles it already carries. Genetic diversity is that hidden reserve — and once it's gone, it's almost impossible to rebuild.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Genetic diversity is when there are lots of different animals and plants in an area. Variation comes from animals breeding. Why it loses marks: describes species diversity, not genetic diversity within a population. "Breeding" recombines existing alleles — it isn't a source of new variation.

✓ STRONG / MODEL ANSWER

Genetic diversity is the variety of alleles in the gene pool of a population or species. A source of new genetic variation is mutation, which creates entirely new alleles. Why it scores: precise allele-level definition + correctly identifies mutation as the source of new alleles (not recombination).

✗ WEAK ANSWER

A population with high genetic diversity is stronger so it can fight off disease better. The animals will adapt to the disease and become immune, then survive and not go extinct. Why it loses marks: "stronger" is vague; "animals adapt to become immune" is the Lamarckian error — individuals don't adapt on demand. No mention of alleles, selection or allele frequency.

✓ STRONG / MODEL ANSWER

High genetic diversity means many different alleles exist in the gene pool. A new disease acts as a selection pressure. Because more alleles are present, it is more likely that some individuals already carry an allele giving resistance. Those individuals survive and reproduce, passing the resistance allele on, so its frequency rises and the population recovers. A low-diversity population may contain no resistant individuals, so it cannot respond and may collapse. Why it scores: alleles → selection pressure → pre-existing resistance → survival & reproduction → frequency shift → contrast with low-diversity outcome. Full causal chain.

✗ WEAK ANSWER

When there are not many birds left there is less diversity because there are fewer of them. This is bad for the birds. Why it loses marks: circular ("fewer = less diversity") with no mechanism (drift/inbreeding). "Bad for the birds" is not a named consequence.

✓ STRONG / MODEL ANSWER

In a small population, mating between related individuals (inbreeding) is unavoidable and genetic drift causes allele frequencies to change randomly, so rare alleles are lost and heterozygosity falls — reducing genetic diversity. A consequence is inbreeding depression: more harmful recessive alleles are expressed, lowering fertility and survival. Why it scores: names two mechanisms (inbreeding, drift) with their effect on alleles, then a specific consequence (inbreeding depression with examples).

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.